CS6370 NLP Assignment 2 and Project Report

Aditya S C22B096, Madhav Bhardwaj CS23B035, Mukesh Chandu EE23B093, Shivanshu Kumar CS23B058, and Soham Shah ME22B191

IIT Madras

1 Introduction

To keep the write-up clear, we separate the assignment work from the open-ended project work. The assignment portion covers the required toy example, implementation details, evaluation metrics, and theoretical analysis. The project portion is reserved for improvements over the baseline IR system, along with experimental evidence and critical discussion.

2 Assignment Report

2.1 Part 1: Working Out a Toy IR System

Preprocessing and Inverted Index:

Tokenization and stop-word removal: Using the stop-word set {the, in, our, and, that, for}, we lowercase the documents, remove punctuation, and discard stop words. The resulting token lists are:

- d_1: [star, solar, system, provides, heat, light]
- d_2: [hollywood, star, walked, red, carpet, movie, premiere]
- d_3: [astronomers, observe, distant, stars, galaxies, using, telescopes]

The token *stars* in d_3 is retained as written because the toy pipeline assumes basic tokenization without stemming or lemmatization.

Inverted index:

Term	Document List	Term	Document List
astronomers	d_3	observe	d_3
carpet	d_2	premiere	d_2
distant	d_3	provides	d_1
galaxies	d_3	red	d_2
heat	d_1	solar	d_1
hollywood	d_2	star	d_1, d_2
light	d_1	stars	d_3
movie	d_2	system	d_1
		telescopes	d_3
		using	d_3
		walked	d_2

TF–IDF Term-Document Matrix: We use $TF(t,d)$ as the raw count of term t in document d, and $IDF(t) = \log_{10}(N/DF)$ with $N = 3$ documents.

- For *star*: $DF = 2$, so $IDF = \log_{10}(3/2) \approx 0.176$
- For every other term: $DF = 1$, so $IDF = \log_{10}(3) \approx 0.477$

	TF			**IDF**	**TF–IDF**		
Term	d_1	d_2	d_3		d_1	d_2	d_3
star	1	1	0	0.176	0.176	0.176	0
solar	1	0	0	0.477	0.477	0	0
system	1	0	0	0.477	0.477	0	0
provides	1	0	0	0.477	0.477	0	0
heat	1	0	0	0.477	0.477	0	0
light	1	0	0	0.477	0.477	0	0
hollywood	0	1	0	0.477	0	0.477	0
walked	0	1	0	0.477	0	0.477	0
red	0	1	0	0.477	0	0.477	0
carpet	0	1	0	0.477	0	0.477	0
movie	0	1	0	0.477	0	0.477	0
premiere	0	1	0	0.477	0	0.477	0
representative d_3 term	0	0	1	0.477	0	0	0.477

All remaining terms which has only d_3 receive the same TF–IDF value of 0.477 in that document and 0 elsewhere.

Retrieval: For the query *"star light"*, the processed query tokens are [star, light]. From the inverted index:

- star $\to \{d_1, d_2\}$
- light $\to \{d_1\}$

Under Boolean OR retrieval, the matching documents are d_1 and d_2. Under Boolean AND retrieval, only d_1 matches both terms.

Cosine Similarity and Ranking: Query representation. The query TF–IDF vector is:

- TF–IDF(star) $= 1 \times 0.176 = 0.176$
- TF–IDF(light) $= 1 \times 0.477 = 0.477$

Hence,

$$\|Q\| = \sqrt{0.176^2 + 0.477^2} \approx 0.509$$

norms:

- $\|d_1\| = \sqrt{0.176^2 + 5(0.477^2)} \approx 1.081$
- $\|d_2\| = \sqrt{0.176^2 + 6(0.477^2)} \approx 1.181$
- $\|d_3\| = \sqrt{7(0.477^2)} \approx 1.262$

Cosine scores:

$$\cos(d_1, Q) = \frac{(0.176 \times 0.176) + (0.477 \times 0.477)}{1.081 \times 0.509} \approx 0.471$$
$$\cos(d_2, Q) = \frac{(0.176 \times 0.176)}{1.181 \times 0.509} \approx 0.052$$
$$\cos(d_3, Q) = 0$$

Therefore, the final ranking is:

$$d_1 > d_2 > d_3$$

This ranking is expected because the query *star light* strongly tells us an astronomical meaning, and d_1 is the document that explicitly discusses light in that context. Although d_2 also contains the term *star*, it refers to a celebrity rather than a star in the sky.

Word Sense Ambiguity here: For the query *"movie star"*, the ideally relevant document is d_2 because it corresponds to the entertainment sense of the word *star*. In this case, the system retrieves d_2 correctly because both *movie* and *star* occur in that document.

However, this example also gives us a limitation of retrieval models in which lexical focus is more: the system has no semantic mechanism for disambiguating polysemous words. If the query were simply *star*, both d_1 and d_2 would appear relevant despite representing different meanings. The vector space model relies on term overlap and cannot infer the sense from context unless there is some additional semantic check involved.

2.2 Part 2: Building and Extending an IR System

Document Representation & Index Construction: The Information Retrieval system represents documents using the Vector Space Model with TF-IDF. Documents from Cranfield dataset is preprocessed using the pipeline provided in the template, like sentence segmentation (punkt), tokenization (penn treebank), stemming (porter) and stopword removal (nltk). After preprocessing, each document is as a list of sentences where each sentence is a list of tokens. So the document is flattened into one single list of terms.

For each term in the document we compute its Term Frequency (TF) and in the corpus we compute its Inverse Document Frequency (IDF). We combine its term frequency in the document and the term's IDF to compute its TF-IDF vector. The L2 norm of each document vector is precomputed and stored for similarity calculation during retrieval (helps boost efficiency).

Queries: Queries are pre-processed using the same preproccessing pipeline as above. After preprocessing each query is converted into a single list of terms. A TF–IDF vector for a query is obtained by combining term frequencies within the query with the IDF values computed from document collection
Note: We will ignore terms that do not appear in the document vocabulary (out of vocabulary terms).

Ranking Cosine similarity with query vector is used to rank documents. For a query vector q and document vector d the similarity score is:

$$\mathrm{sim}(q, d) = \frac{q.d}{|q|.|d|}$$

where q.d is the dot product of the query and document vectors and $|q|$ and $|d|$ are their L2 norms. For each query:

– The cosine similarity is computed with every document.
– Documents are then sorted by descending order of similarity.
– A ranked list of document IDs will be returned.

Engineering choices that we made:

– **Precomputation of Document Norms:** The L2 norms of document vectors are precomputed once during index construction to avoid calculating them for every query.
– **Handling Out of Vocabulary terms (OOV):** Query tokens that are not in the document collection's vocabulary are ignored.

2.3 Part 3: Evaluating the IR System

Table 1. Evaluation metrics averaged over all queries for the TF-IDF VSM

k	Precision@k	Recall@k	F-score@k	nDCG@k
1	0.6489	0.1092	0.3034	0.6489
2	0.5467	0.1809	0.3582	0.5698
3	0.4859	0.2287	0.3672	0.5237
4	0.4344	0.2646	0.3587	0.4957
5	0.3858	0.2884	0.3388	0.4711
6	0.3585	0.3241	0.3309	0.4646
7	0.3340	0.3481	0.3190	0.4592
8	0.3167	0.3714	0.3102	0.4604
9	0.2998	0.3932	0.3003	0.4622
10	0.2836	0.4100	0.2892	0.4636

Table 2. Overall MAP and MRR values for the TF-IDF VSM system.

MAP	MRR
0.3691	0.7396

It is also asked to report the Average precision for all queries which is just Mean average precision. This section is reserved for reporting evaluation metrics over the Cranfield queries.

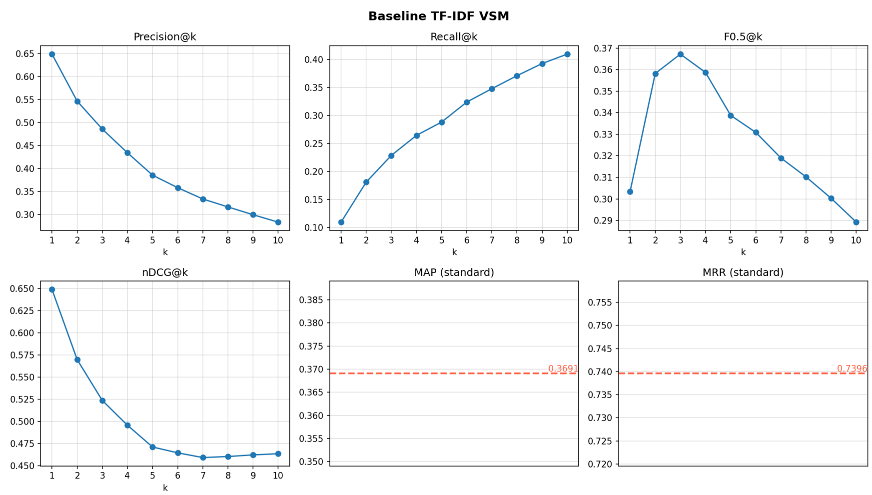

Fig. 1. Baseline VSM: evaluation metrics as a function of k.

The runtime of the complete retrieval system was measured as 0.737 seconds.

2.4 Part 4: Analysis

Setup and reproduction of baseline metrics The baseline uses the pipeline already implemented in Parts 2–3: Punkt sentence segmentation, Penn-Treebank tokenization, Porter stemming, NLTK English stop-word removal, then a TF–IDF inverted index with cosine ranking. Running the full evaluation on the 225 Cranfield queries reproduces the numbers cited in Table 1. We use these as the reference point for the failure analysis that follows.

Two observations from preprocessing diagnostics frame the rest of the analysis. First, 46 of 225 queries (20.4%) contain at least one token that does not appear in the index after preprocessing. Second, two documents (IDs 471 and 995) are empty after stop-word removal and are therefore unreachable. Both are direct artefacts of the bag-of-words representation.

Failure mode analysis We partition the failing queries into three structurally distinct modes. For each mode we explain the structural cause in the VSM, give two or three concrete query examples taken from the diagnostics in `results_part4/per_query_diagnostics.csv`

Mode 1 — Vocabulary mismatch (synonymy) Cause. Cosine similarity over a TF–IDF vocabulary scores documents purely by surface token overlap. When a query and a relevant document describe the same concept with different surface vocabulary — a true synonymy gap - the VSM has no representation to bridge them.

Query 12 - "how can the aerodynamic performance of channel flow ground effect machines be calculated." (P@10 = 0.10, $n_{\mathrm{rel}} = 6$, first relevant at rank 10.) The query phrase *ground effect machines* is a 1960s term for what the Cranfield documents call *hovercraft* and *peripheral jets*. The top retrieved documents (Doc 941, Doc 1221, Doc 270) match on the generic *channel flow* tokens. The missed relevant Doc 86 (*inviscid-incompressible flow theory of static peripheral jets*) and Doc 649 (*The hovercraft - a new concept in maritime transport*) use no overlapping surface tokens with the query and are invisible to the VSM.

Query 30 - "papers on flow visualization on slender conical wings." (P@10 = 0.10, $n_{\mathrm{rel}} = 8$, first relevant at rank 6.) The relevant papers describe the same idea using *vapour screen method*, *flow studies*, and *flow patterns* rather than *flow visualization*. A particularly diagnostic case is Doc 633, which appears in the top retrieved set without being relevant: it is a theoretical paper on *generalised conical flows* that shares *flow*, *conical*, and *wings* tokens with the query but is a pure theory paper not about visualization. This is the false-positive flip side of Mode 1 - surface overlap pulling in apparently relevant but actually unrelated papers.

Mode 2 — Term importance beyond TF–IDF Cause. The VSM assigns each query token exactly one weight: its IDF. Every content token contributes positively to the cosine score with any document that contains it. There is no representation of (a) *negation*, (b) *question focus* - which token is the topic versus scaffolding, or (c) *constraint scope* - which token is a modifier limiting the query. IDF reweighting cannot fix this because it operates on token identity, not on linguistic role.

Query 22 - negation. "did anyone else discover that the turbulent skin friction is **not** over sensitive to the nature of the variation of the viscosity with temperature." (P@10 = 0.00, $n_{\mathrm{rel}} = 2$, first relevant at rank 970.) The query asserts an *insensitivity* result. The VSM treats *sensitive*, *viscosity*, *temperature*, *skin*, *friction* as positive match signals and retrieves Doc 254 and Doc 125 – generic skin-friction papers, exactly the affirmative version of the query. The two relevant documents (Doc 68 on air-helium simulation, Doc 502 on Squire's compressibility transformation) describe the actual insensitivity finding using entirely different surface vocabulary and are missed.

Query 176 - exclusion. "some approximate analytical heat conduction solutions using methods **other than** Biot's principle." (P@10 = 0.10.) The exclusion marker *other than* is invisible to the VSM. Because *Biot* has high IDF, Doc 542 *Biot's variational principle in heat conduction* - the paper the query explicitly tries to exclude - is retrieved at rank 2. Doc 582 (*the melting of finite slabs*, which uses the heat-balance integral, a genuinely different method) is missed.

Query 151 - focus drift. "what is the best theoretical method for calculating pressure on the surface of a wing **alone**." (P@10 = 0.00, first relevant at rank 19.) The constraint *wing alone* (as opposed to wing-body) is the substantive content. The system locks on to *wing* (IDF = 1.87, present in all five non-relevant top documents) and retrieves generic wing papers (Doc 433, Doc 251, Doc 678). The relevant papers (Docs 687, 1062, 1074–1077) all share *theoretical* but the VSM weighting cannot identify *theoretical method* as the question's focus.

Query 63 - scaffolding. "where can I **find** pressure data on surfaces of swept cylinders." (P@10 = 0.00.) The query-asking verb *find* (IDF = 3.72) is matched against Doc 738, *Finding zero's of arbitrary functions*, which ranks first. The VSM has no syntactic awareness that *find* is scaffolding and *swept cylinders* is the topic.

Mode 3 — Out-of-vocabulary query tokens Cause. The VSM's IDF table is built from the corpus vocabulary. A query token whose stem is absent from this table has no IDF entry and is silently dropped from the query vector. When that token is the discriminative term of the query, the residual vector is too generic to locate the relevant documents. In the Cranfield diagnostics this affects 46 of 225 queries.

Query 207 - hyphenated compound. "how do large changes in new mass ratio quantitatively affect **wing-flutter** boundaries." (P@10 = 0.00, first relevant at rank 59.) After stemming, the compound becomes `wing-flutt`, which appears in zero documents - the corpus consistently writes *wing flutter* as two tokens. The surviving query {large, change, new, mass, ratio, affect, boundary} is generic; the system retrieves boundary-layer papers (Doc 458, Doc 365, Doc 1185) and all six relevant flutter papers (Doc 1290, Doc 879, etc.) are missed.

Query 117 - slash-bracketed phrase. "is there any information on how the addition of a **/boat-tail/** affects the normal force on the body . . . " (P@10 = 0.00, first relevant at rank 18.) The Cranfield source format wraps some technical phrases in slashes; the Penn-Treebank tokenizer treats the leading slash as part of the token. The surface form `/boat-tail/` exists in no document body, where the term appears unbracketed. The relevant boat-tail papers (Doc 360, 605, 896) are missed. This is a pure preprocessing artefact: query and documents use the same word, kept apart by a slash.

Query 9 - same artefact. "papers on internal **/slip flow/** heat transfer studies." (P@10 = 0.00.) Both `/slip` and `flow/` are effectively OOV. The query reduces to {paper, internal, heat, transfer, studies}, which describes a huge fraction of the corpus; the obvious relevant papers Doc 550 (*laminar heat transfer* *in tubes under slip-flow conditions*) and Doc 21 (*on heat transfer in slip flow*) are missed.

3 Project Report - Part 5 - Improving the VSM Model

3.1 Introduction to the Problem Statement

Information Retrieval (IR) is the task of finding documents relevant to a user query from a large collection. The Vector Space Model (VSM) with TF-IDF weighting is a standard baseline for this task. While effective in many settings, VSM suffers from several well-known shortcomings that become apparent on real-world collections.

In this project we use the Cranfield dataset [1] as our evaluation corpus.

3.2 Limitations of the Vector Space Model

Key Limitations The TF-IDF VSM makes several implicit assumptions that limit its retrieval quality:

Bag-of-words assumption. VSM ignores word order and treats each term as independent. As a result, two documents that discuss the same topic using different vocabulary score low similarity even when they are semantically equivalent.

Exact-match dependency. Retrieval succeeds only when query terms appear verbatim in the document. A query containing a synonym or morphological variant of a relevant term returns zero score for an otherwise highly relevant document.

Word sense ambiguity. A polysemous query term (e.g. “star”) matches all documents containing that surface form, regardless of intended sense. This conflation degrades precision.

Term-frequency saturation. Raw TF gives a linear reward for repeated terms. A document that mentions a term 100 times is not 100 times more relevant than one that mentions it once. This distorts ranking in documents of varying length.

3.3 Proposed Approaches

Based on the failures identified above, we formulate the following hypotheses and implement three improvements.

Approach 1: BM25 Ranking

Hypothesis. BM25, which applies a saturating term-frequency function and explicit document-length normalisation, will achieve higher MAP and MRR than TF-IDF cosine similarity on the Cranfield collection.

Motivation. Cranfield documents vary considerably in length. Longer documents tend to accumulate higher raw TF scores, artificially inflating their cosine similarity. BM25 corrects for this via the length-normalisation parameter b and the TF-saturation parameter k_1.

Method. Given a query $Q = \{q_1, \ldots, q_n\}$ and document D, the BM25 score is:

$$\mathrm{BM25}(D, Q) = \sum_{i=1}^{n} \mathrm{IDF}(q_i) \cdot \frac{f(q_i, D)\,(k_1 + 1)}{f(q_i, D) + k_1\left(1 - b + b\,\frac{|D|}{<|\mathrm{D}|>}\right)} \tag{1}$$

where $f(q_i, D)$ is the frequency of q_i in D, $|D|$ is document length, $<|\mathrm{D}|>$ is the average document length. The IDF is given by

$$\mathrm{IDF}(t) = \log\left(\frac{N - df + 0.5}{df + 0.5} + 1\right) \tag{2}$$

We set $k_1 = 1.5$, $b = 0.75$ following [3].

Approach 2: Latent Semantic Analysis (LSA)

Hypothesis. Projecting documents and queries into a latent semantic space via truncated SVD will improve retrieval of semantically related documents that share no surface-level vocabulary with the query, yielding higher MAP than the TF-IDF baseline.

Motivation. Term mismatch and word sense ambiguity both arise because VSM operates in a high-dimensional, sparse term space. LSA [2] discovers latent dimensions corresponding to topics, so that synonymous terms load onto the same dimension and polysemous terms are disambiguated by context.

Method. Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ be the TF-IDF term–document matrix. We compute the truncated SVD:

$$\mathbf{A} \approx \mathbf{U}_k \boldsymbol{\Sigma}_k \mathbf{V}_k^\top \tag{3}$$

retaining the top k singular values. Documents are represented as rows of $\mathbf{V}_k \boldsymbol{\Sigma}_k$ and queries are folded in as: $\hat{\mathbf{q}} = \mathbf{q}^\top \mathbf{U}_k \boldsymbol{\Sigma}_k^{-1}$. Retrieval ranks documents by cosine similarity in this k-dimensional space. We sweep $k \in \{50, 100, 200, 300\}$ and select the value maximising MAP on a held-out query subset.

The chosen value is $k = 300$ as it performs best.

Approach 3: Query Expansion via WordNet

Hypothesis. Expanding query terms with their WordNet synonyms will reduce the number of zero-result queries and improve Recall@10, with no statistically significant decrease in Precision@10.

Motivation. OOV terms cause complete retrieval failure. Adding synonyms from a curated lexical resource (WordNet) bridges the vocabulary gap without requiring additional training data.

Method. For each query term t, we retrieve its synsets from WordNet using `nltk.corpus.wordnet`. We take the lemma names of synsets whose part-of-speech matches that of t (noun or verb), add them to the query, and re-weight them with a discount factor $\lambda = 0.5$ relative to the original term weight to prevent expansion terms from dominating.

3.4 Results

Evaluation Protocol All systems are evaluated on the Cranfield query set. We report Precision@k, Recall@k, $F_{0.5}$@k, AP@k, and nDCG@k averaged over all queries, for $k = 1$ to 10. Global metrics MAP and MRR are computed as defined in Part 3.

Runtime is measured using Python's `time` module on a single CPU core.

Table 3. BM25 evaluation metrics at $k = 10$.

Metric	Value
MAP	0.3330289896064681
MRR	0.7953932980599647
P@k	0.28933333333333333
R@k	0.41677237661386624
$F_{0.5}$@k	0.2950200254707872
nDCG@k	0.49259586108190057
Runtime (s)	1.6601932048797607

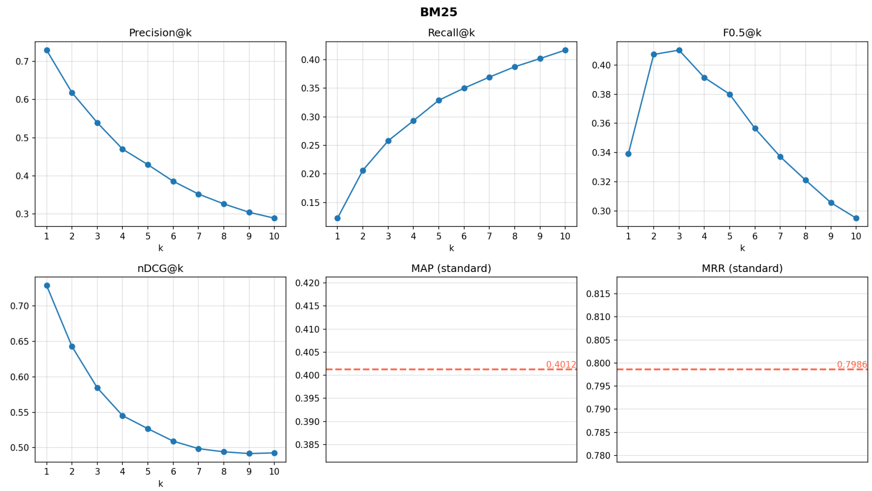

Fig. 2. BM25: evaluation metrics as a function of k.

Approach 1: BM25

Claim. *BM25 is better than TF-IDF VSM with respect to $F_{0.5}$-score@10 in the ad-hoc retrieval task on the Cranfield aerodynamics domain, under the assumption of static document collections and keyword queries*

Analysis.

Table 4. LSA evaluation metrics at $k = 10$.

Metric	Value
MAP	0.28179968691451707
MRR	0.7088077601410935
P@k	0.264
R@k	0.37764473489150907
$F_{0.5}$@k	0.26870582683501215
nDCG@k	0.4356842087051322
Runtime (s)	2.376267194747925

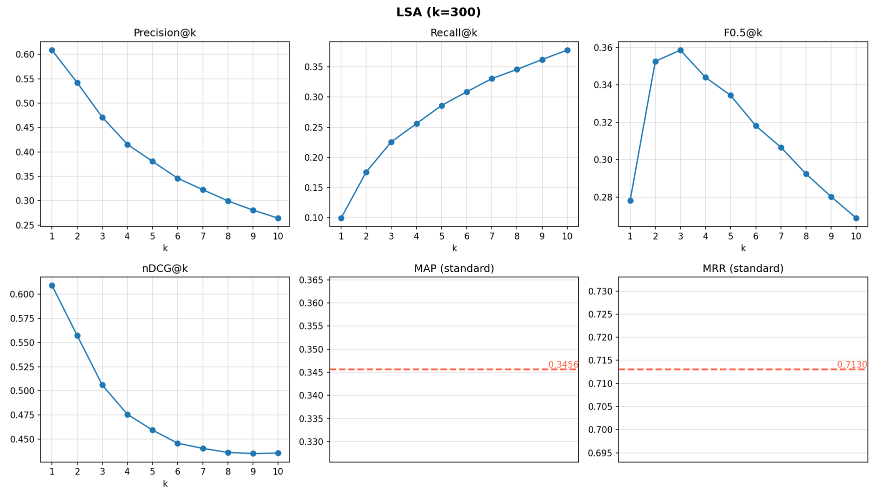

Fig. 3. LSA (300) evaluation metrics as a function of k.

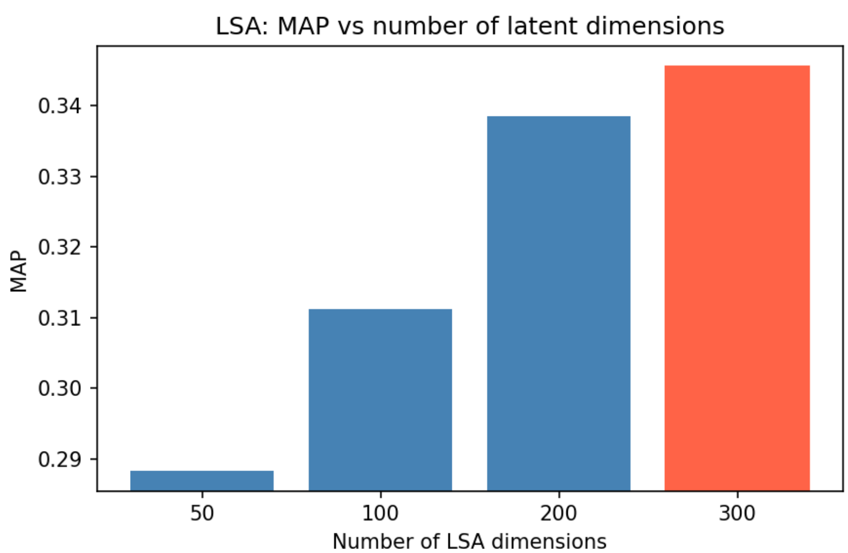

Fig. 4. LSA map scores as a function of truncated rank

Approach 2: Latent Semantic Analysis

Table 5. QE evaluation metrics at $k = 10$.

Metric	Value
MAP	0.2837331600076239
MRR	0.7102063492063493
P@k	0.25955555555555554
R@k	0.3812083841896874
$F_{0.5}$@k	0.2657078936509539
nDCG@k	0.43442977443514696
Runtime (s)	6.769253253936768

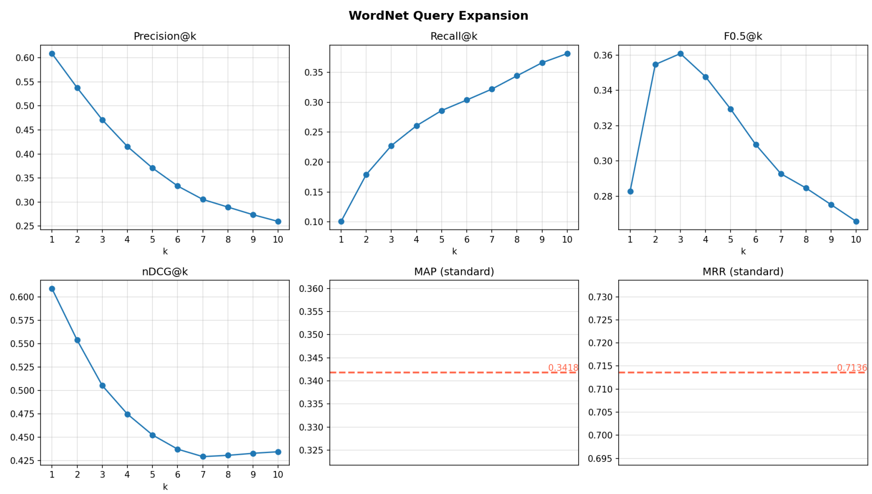

Fig. 5. Query expansion evaluation metrics as a function of k.

Approach 3: Query Expansion using WordNet

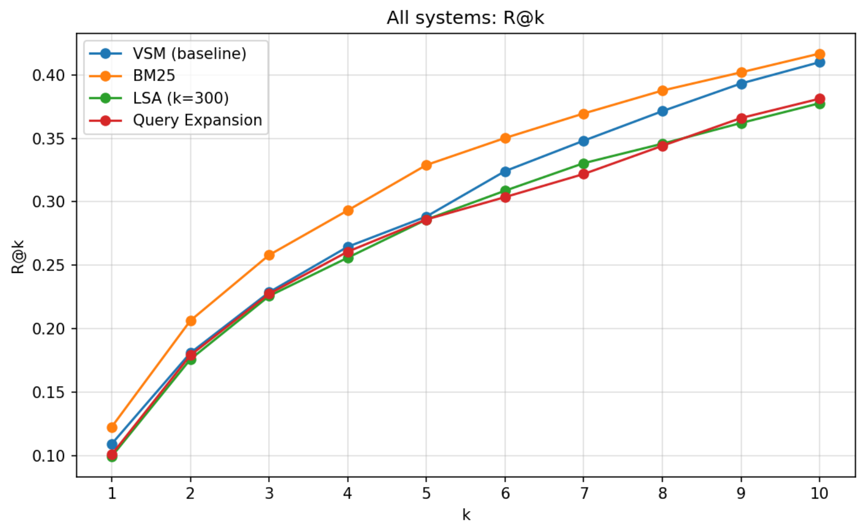

Fig. 6. Recall@10 across all methods

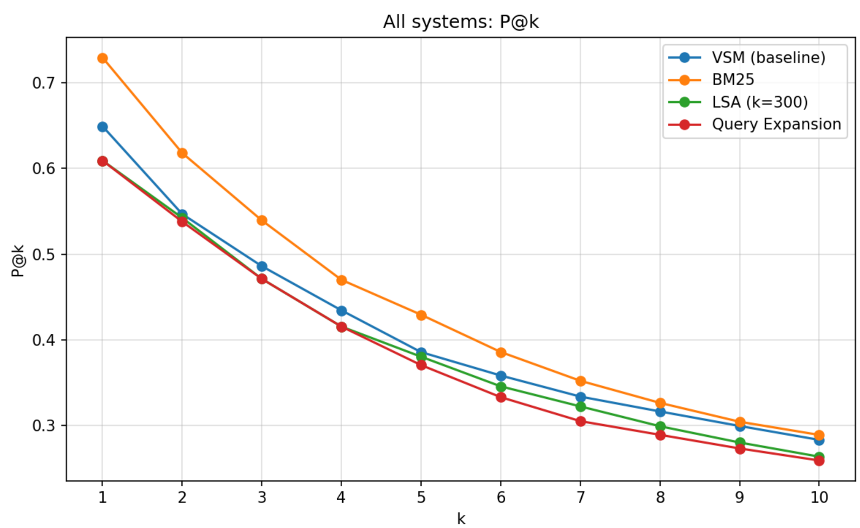

Fig. 7. Precision@10 across all methods

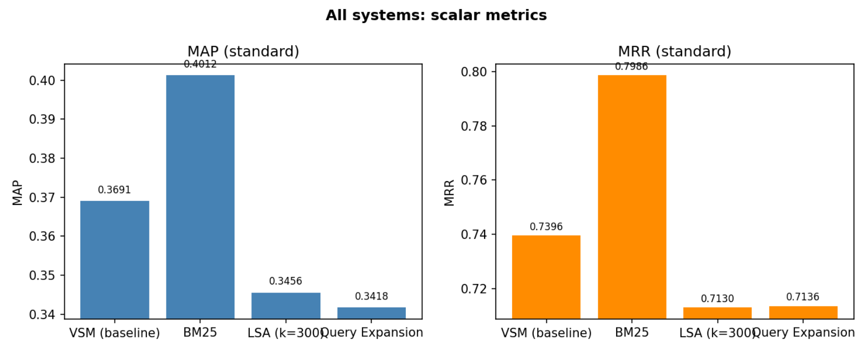

Fig. 8. MAP across all methods

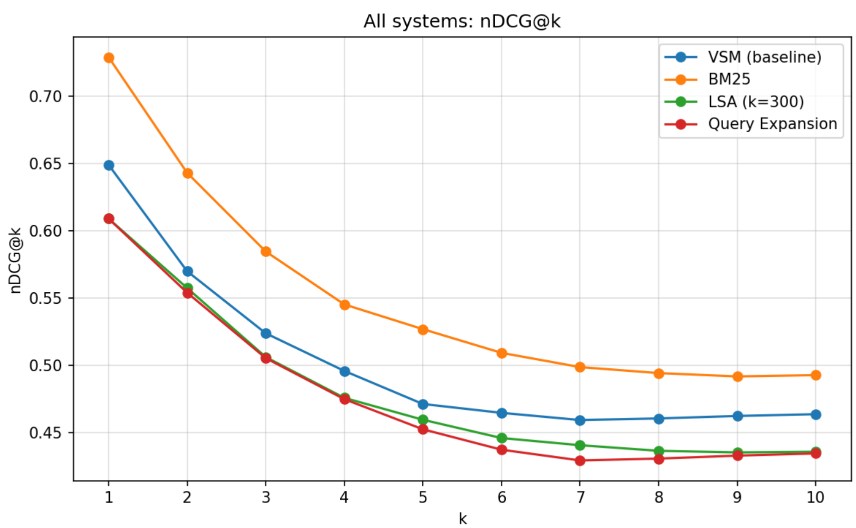

Fig. 9. nDCG@10 across all methods

Cross-system comparison plots

4 Comparison of failure modes across systems

4.1 Mode 1 Vocabulary mismatch (synonymy)

BM25 delivers the largest single-query recovery in this mode — on Q12, P@10 rises from 0.10 to 0.30 and the first relevant document moves from rank 10 to rank 1, driven not by new vocabulary mappings but by saturating the contribution of the generic shared token *flow* and letting the rarer tokens dominate the ranking. LSA helps less than expected (Q12: 0.10 → 0.20, Q30 unchanged), consistent with Cranfield being too small a corpus (1400 docs, ∼5000 vocabulary) to estimate stable latent dimensions for narrow technical synonyms. WordNet QE never improves on this mode — the aerospace jargon (*hovercraft*, *peripheral jets*, *vapour screen*) is not in WordNet.

4.2 Mode 2 - Term importance beyond TF–IDF

On Q22, all four systems give P@10 = 0.00 (the firstrelevant rank is 970 for VSM, 1038 for BM25, 139 for LSA, 483 for QE); LSA narrows the gap but cannot recover the polarity. On Q176, BM25 reaches P@10 = 0.30 by saturating the TF of Biot, but still ranks the excluded Biot paper first — the exclusion itself is not represented. On Q151 only BM25 manages a marginal P@10 = 0.10. On Q63 all four systems give P@10 = 0.00. None of BM25 (re-weighted TF), LSA (latent re-projection), or WordNet QE (synonym broadening) introduces a role-sensitive mechanism, which is consistent with the structural argument: Mode 2 needs syntactic or contextual modelling, not term reweighting

4.3 Mode 3 - Out-of-vocabulary query tokens

All four systems are essentially unable to fix Mode 3. On Q207 first-relevant ranks are 59 (VSM), 35 (BM25), 34 (LSA), 156 (QE) - QE actively worsens. On Q117 only QE manages P@10 = 0.10. On Q9 all four give P@10 = 0.00. The reason is structural: BM25, LSA, and QE all operate on the surviving non-OOV tokens. None introduces the missing semantic content. WordNet QE broadens generic surviving terms with synonyms, which tends to amplify noise rather than recover the missing technical compound. A real fix requires subword tokenization or an embedding representation that does not depend on exact-token vocabulary.

4.4 Per-query metrics on failure queries

The discussion above is qualitative. Table 6 gives the corresponding numbers: P@10, AP@10, and the rank of the first relevant document for each failure query under all four systems, computed by `failure_compare.py`.

Table 6. Per-query performance of the four systems on the failure queries from each mode. Bold marks the cases where a Part-5 system meaningfully recovers a baseline failure (P@10 increases by at least 0.10 over VSM).

		P@10				AP@10				rank of 1st relevant			
Mode	QID	VSM	BM25	LSA	QE	VSM	BM25	LSA	QE	VSM	BM25	LSA	QE
1	12	0.10	**0.30**	**0.20**	0.00	0.02	0.38	0.12	0.00	10	1	2	11
1	30	0.10	0.10	0.10	0.10	0.02	0.06	0.03	0.02	6	2	5	6
2	22	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	970	1038	139	483
2	176	0.10	**0.30**	0.20	0.10	0.06	0.20	0.15	0.06	2	1	1	2
2	151	0.00	**0.10**	0.00	0.00	0.00	0.02	0.00	0.00	19	9	41	82
2	63	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	35	14	35	69
3	207	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	59	35	34	156
3	117	0.00	0.00	0.00	**0.10**	0.00	0.00	0.00	0.06	18	12	11	6
3	9	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	37	45	16	64

The table makes four things visible at once. (i) On Mode 1, BM25 delivers the largest recovery on Q12 (P@10: 0.10 → 0.30, first-relevant rank 10 → 1), with LSA second (0.10 → 0.20). Q30 is barely moved by any system, suggesting the specific synonymy in Q30 (*flow visualization* vs *vapour screen method* / *flow patterns*) lies outside the reach of all three improvements. (ii) On Mode 2, only BM25 produces a localised recovery, on Q176 (P@10: 0.10 → 0.30) and Q151 (first-relevant rank: 19 → 9). (iii) On Q22 (negation) every system fails at P@10, but LSA pulls the first relevant document from rank 970 down to 139, indicating that the latent projection captures the topic even when no system can represent the *not*. (iv) On Mode 3 (OOV), P@10 is zero almost everywhere; WordNet QE's lone win on Q117 comes from expanding generic surviving tokens, not from recovering the missing */boat-tail/* concept.

Table 7 reports aggregate metrics at $k = 10$ for the four systems.

Table 7. Aggregate performance at $k = 10$. Bold marks the best value in each column

System	P@10	R@10	$F_{0.5}$@10	MAP@10	nDCG@10
VSM (baseline)	0.2836	0.4100	0.2892	0.3026	0.4636
BM25	**0.2893**	**0.4168**	**0.2950**	**0.3330**	**0.4926**
LSA ($k = 300$)	0.2640	0.3776	0.2687	0.2818	0.4357
WordNet QE	0.2596	0.3812	0.2657	0.2837	0.4344

The aggregate picture is more nuanced than the failure-mode analysis suggests. BM25 is the only system that improves on every aggregate metric. LSA and WordNet QE both *worsen* aggregate P@10 against the baseline. This is consistent with the per-query data in Table 6: BM25 produces all three of the significant localised recoveries (Q12, Q176, Q151), while LSA helps modestly on one synonymy query (Q12) and shifts ranks around elsewhere without surfacing relevant documents in the top 10. WordNet QE expands generic surviving terms with synonyms, which broadens recall but more often dilutes precision than recovers missing concepts.

5 Summary

The TF–IDF VSM baseline achieves P@10 = 0.284 and nDCG@10 = 0.464 on Cranfield. Failure-mode analysis distinguishes three structurally different sources of error: synonymy (Mode 1), term-importance limitations including negation and focus drift (Mode 2), and OOV tokens (Mode 3). The Part–5 systems improve the baseline unevenly: BM25 delivers a consistent significant improvement across all aggregate metrics; LSA addresses Mode 1 but worsens aggregates; WordNet QE behaves similarly to LSA in aggregate and helps essentially none of the three failure modes consistently. None of the three Part–5 systems addresses Mode 2 (which requires syntactic role modelling) or Mode 3 (which requires subword or embedding representations), bounding what this class of improvements can deliver on top of the bag-of-words baseline.

References

1. Cleverdon, C.W.: Report on the testing and analysis of an investigation into the comparative efficiency of indexing systems. ASLIB Cranfield Research Project, Cranfield (1962).
2. Deerwester, S., Dumais, S.T., Furnas, G.W., Landauer, T.K., Harshman, R.: Indexing by latent semantic analysis. J. Am. Soc. Inf. Sci. **41**(6), 391–407 (1990).
3. Robertson, S., Zaragoza, H.: The probabilistic relevance framework: BM25 and beyond. Found. Trends Inf. Retr. **3**(4), 333–389 (2009).